Chemistry Topic 3 — Metals, Non-Metals and Chemical Reactions

Covers syllabus (click to expand)

Metals & Non-Metals Physical & Chemical Tags | Allotropes of Carbon (Graphite, Diamond, Fullerenes, Graphene) & Thermodynamic Stability | Metal Reactivity Series & Single Displacement Dynamics | Reaction with Water, Acids & Aqua Regia | Important Ores & Extractive Metallurgy (Bayer, Hall-Héroult, Blast Furnace, Froth Flotation) | Master Alloy Systems (Brass, Bronze, German Silver, Gun Metal, Solder, Stainless Steel, Nichrome, Amalgams) | Types of Chemical Reactions (Combination, Decomposition, Displacement, Double Displacement) | Redox Chemistry (Oxidation Numbers, Electron Transfer, Oxidizing/Reducing Agents) | Corrosion Mechanisms (Rusting of Fe, Patina on Cu, Tarnish on Ag) & Protective Technologies (Galvanization, Anodizing, Sacrificial Anodes)

Sources baked in: NCERT Class 8, 9 & 10 (Metals and Non-metals; Chemical Reactions and Equations), Lucent General Science (Chemistry), Physics Wallah (PW) State PCS Foundation, Ghatna Chakra General Science (1990–2025 PYQs). **Weight:** ★★★ High — Highly recurring questions on German Silver composition (0% Ag), Galvanization (Zinc coating), Amalgam exceptions (Iron does not form amalgams), Graphite stability vs Diamond hardness, Aqua Regia ratio (3:1 HCl:HNO3), and Ore identification (Cinnabar, Bauxite, Pitchblende, Galena). **Last verified:** September 2026

Current Affairs (2024–2026)

Year	Event	Core Concept & Must-Score Significance
2025–2026	**Critical Minerals Mission & Lithium Reserves**	Geological Survey of India (GSI) exploration of **Lithium** reserves in Reasi (Jammu & Kashmir) and Degana (Rajasthan). Lithium (Li, Z=3) is the lightest metal and lowest-density solid element (0.534\text{ g/cm}^3), with the highest negative standard reduction potential (-3.04\text{ V}), making it the bedrock of EV lithium-ion batteries.
2024	**UPPCS Prelims 2024**	Question on **German Silver composition**: Highlighted the recurring trap that German Silver contains **0\%\text{ Silver}** (Cu \approx 50\%, Zn \approx 30\%, Ni \approx 20\%). Named solely for its silvery-white metallic luster.
2023	**Noble Prize in Chemistry & Carbon Materials**	Advancements in **Graphene** (single atomic layer of sp^2 carbon) and Carbon Nanotubes (CNTs) for supercapacitors and aerospace composites: Graphene exhibits ballistic electrical conductivity and tensile strength >100 times higher than structural steel.
2022	**UPPCS Prelims 2022**	Core thermodynamic trap: **Graphite** is the thermodynamically most stable allotropic form of carbon at standard temperature and pressure (298.15\text{ K}, 1\text{ bar}), with standard enthalpy of formation \Delta_f H^\circ = 0\text{ kJ/mol} (Diamond has \Delta_f H^\circ = +1.9\text{ kJ/mol}).

Year	Event	Core Concept & Must-Score Significance
2020	**UPPCS Prelims 2020**	Characteristic matching of metals: Sodium (soft, cut with a knife), Mercury (liquid metal), Silver (highest electrical conductivity), and Lead (poorest thermal conductor among common commercial metals).

📋 **Master Reference Tables (Click to Expand)**

Table 1: Master Ore & Metallurgy Classification (NCERT / Lucent Standard) | Target Metal | Principal Mineral Ores & Chemical Formulas | Chemical Nature of Ore | Extraction & Reduction Technology | |---|---|---|---| | **Aluminium (Al)** | **Bauxite** ($Al_2O_3 \cdot 2H_2O$), **Cryolite** (Na_3AlF_6), **Corundum** (Al_2O_3) | Hydrated Oxide / Fluoride | Bayer's process leaching followed by Hall-Héroult electrolytic reduction in molten cryolite/fluorspar. | | **Iron (Fe)** | **Haematite** (Fe_2O_3), **Magnetite** (Fe_3O_4), **Siderite** ($FeCO_3$), **Iron Pyrites** (FeS_2) | Oxide / Carbonate / Sulphide | Blast furnace smelting using coke (reducing agent) and limestone ($CaCO_3$, flux to remove SiO_2 as slag $CaSiO_3$). | | **Copper (Cu)** | **Copper Pyrites** ($CuFeS_2$), **Malachite** ($CuCO_3 \cdot Cu(OH)_2$), **Cuprite** (Cu_2O) | Sulphide / Basic Carbonate | Froth flotation concentration, reverberatory smelting, Bessemerization, and electrolytic refining. | | **Lead (Pb)** | **Galena** (PbS), **Cerussite** ($PbCO_3$), **Anglesite** ($PbSO_4$) | Sulphide / Carbonate | Froth flotation, roasting into PbO, followed by carbon reduction or self-reduction. | | **Zinc (Zn)** | **Zinc Blende / Sphalerite** (ZnS), **Calamine** ($ZnCO_3$), **Zincite** (ZnO) | Sulphide / Carbonate / Oxide | Roasting to ZnO, followed by reduction with crushed coke in vertical retorts at 1200°C. | | **Mercury (Hg)** | **Cinnabar** (HgS) | Sulphide | Direct roasting in air ($HgS + O_2 \to Hg \uparrow + SO_2$); mercury vapors condense directly to liquid. | | **Uranium (U)** | **Pitchblende** (U_3O_8), **Carnotite** ($K_2(UO_2)_2(VO_4)_2 \cdot 3H_2O$) | Complex Oxide / Vanadate | Leaching with sulphuric acid, solvent extraction, precipitation as "yellowcake" (U_3O_8). | | **Thorium (Th)** | **Monazite sand** (phosphate of thorium, cerium, lanthanum) | Phosphate placer mineral | Acid digestion followed by fractional precipitation; abundant in Kerala coastal sands. | | **Silver (Ag)** | **Argentite / Silver Glance** (Ag_2S), **Horn Silver** ($AgCl$) | Sulphide / Halide | MacArthur-Forrest cyanide leaching process: complexation with NaCN followed by zinc displacement. | | **Magnesium (Mg)** | **Carnallite** ($KCl \cdot MgCl_2 \cdot 6H_2O$), **Dolomite** ($CaCO_3 \cdot MgCO_3$), **Magnesite** ($MgCO_3$) | Halide / Carbonate | Dow process: electrolytic reduction of molten anhydrous $MgCl_2$. | --- ### Table 1B: Complete Ghatna Chakra Master Matrix of Minerals, Ores & Chemical Formulas | Metal | Major Minerals & Ores | Chemical Composition / Formula | Must-Score Takeaway | | :--- | :--- | :--- | :--- | | **Sodium (Na)** | Rock Salt
Chilli Saltpetre
Trona
Borax
Glauber's Salt | $NaCl$
$NaNO_3$
$Na_3H(CO_3)_2 \cdot 2H_2O$
$Na_2B_4O_7 \cdot 10H_2O$
$Na_2SO_4 \cdot 10H_2O$ | Stored under kerosene; burns with brilliant golden yellow flame in street lamps; highly vigorous exothermic reaction with cold water. | | **Potassium (K)** | Sylvine
Carnallite |

Schoenite (Picromerite) | KCl
KCl \cdot MgCl_2 \cdot 6H_2O
K_2Mg(SO_4)_2 \cdot 6H_2O | Most reactive metal in standard series; lower ionization energy than sodium; reacts violently with cold water. | | **Magnesium (Mg)** | Magnesite
Dolomite
Carnallite
Kieserite
Epsom Salt (Epsomite) | MgCO_3
CaMg(CO_3)_2
KCl \cdot MgCl_2 \cdot 6H_2O
MgSO_4 \cdot H_2O
MgSO_4 \cdot 7H_2O | Essential central element in chlorophyll porphyrin ring; does not react with cold water, but reacts vigorously with hot water. | | **Calcium (Ca)** | Limestone / Calcite
Gypsum
Fluorspar
Plaster of Paris | CaCO_3
CaSO_4 \cdot 2H_2O
CaF_2
CaSO_4 \cdot \frac{1}{2}H_2O | Gypsum calcined at 120^\circ\text{C} yields Plaster of Paris; limestone calcined in kiln liberates process CO_2 (>5\% global emissions). | | **Aluminium (Al)** | Bauxite
Cryolite
Corundum
Diaspore | Al_2O_3 \cdot 2H_2O
Na_3AlF_6
Al_2O_3
AlO(OH) | Most abundant metal in Earth's crust (8.13\%); Hall-Héroult electrolytic reduction; Cryolite lowers melting point of alumina bath. | | **Copper (Cu)** | Chalcopyrite (Copper Pyrite)
Chalcocite (Copper Glance)
Cuprite (Ruby Copper)
Malachite (Green)
Azurite (Blue) | CuFeS_2
Cu_2S
Cu_2O
Cu_2(CO_3)(OH)_2
Cu_3(CO_3)_2(OH)_2 | Extracted via smelting & Bessemerization; brass utensils tinned (Sn) to prevent copper toxicity; green patina is basic copper carbonate. | | **Iron (Fe)** | Haematite (Red)
Magnetite (Black/Loadstone)
Limonite (Brown)
Siderite (Spathic Iron)
Iron Pyrites (Fool's Gold) | Fe_2O_3 (\sim 69.9\%\text{ Fe})
Fe_3O_4 (\sim 72.4\%\text{ Fe})
FeO(OH) \cdot nH_2O
FeCO_3
FeS_2 | Bailadila (Chhattisgarh) mines world-class haematite; rusting increases weight due to

hydrated iron(III) oxide formation. | | **Lead (Pb)** | Galena
Cerussite
Matlockite | PbS
$PbCO_3$
PbFCl | Does not react with water or steam; poorest conductor of heat along with mercury (34.7 W/mK). | | **Zinc (Zn)** | Zinc Blende (Black Jack)
Smithsonite (Calamine)
Zincite (Red Zinc)
Franklinite | ZnS
$ZnCO_3$
ZnO
$(Fe, Mn, Zn)(Fe, Mn)_2O_4$ | Used for sacrificial galvanization of iron pipes; cannot store edible oils in galvanized pots due to toxic zinc salt formation and rancidity. | | **Manganese (Mn)** | Pyrolusite
Manganite
Hausmannite | MnO_2
MnO(OH)
Mn_3O_4 | Depolarizer in Leclanché dry cells; essential in ferro-manganese alloy steels for wear resistance. | | **Mercury (Hg)** | Cinnabar | HgS | Only metal liquid at room temperature (25°C); high surface tension; high thermal expansion uniformity; amalgam formation. | | **Silver (Ag)** | Native Silver
Argentite (Silver Glance)
Cerargyrite (Horn Silver) | Ag
Ag_2S
AgCl | Best electrical and thermal conductor of all metals; black tarnishing in air caused by traces of hydrogen sulfide ($2Ag + H_2S \to Ag_2S + H_2$). | | **Tin (Sn)** | Cassiterite (Tinstone) | SnO_2 | Used to plate food cans because tin is less reactive than zinc and resists organic food acids; constituent of Bronze (Cu+Sn) and Solder (Pb+Sn). | | **Titanium (Ti)** | Ilmenite
Rutile | $FeTiO_3$
TiO_2 | Strategic beach placer sand mineral along Kerala, Tamil Nadu, and Odisha coastlines; high strength-to-weight ratio. | | **Thorium (Th)** | Monazite | Phosphate of rare earths & Thorium (ThO_2) | Heavy beach placer sand mineral found in Kerala, Tamil Nadu, and Odisha; cornerstone of India's three-stage nuclear power program. | ### Table 2: Commercial & Strategic Alloys Matrix | Alloy Name | Nominal Composition (% by weight) | Special Engineering Properties | Core Practical & Industrial Applications | |---|---|---|---| | **Brass** | $Cu\text{ (60–70\%)}$, $Zn\text{ (30–40\%)}$ | Malleable, acoustic resonance, corrosion-resistant | Musical brass instruments, ammunition cartridge cases, plumbing valves, decorative hardware | | **Bronze** | $Cu\text{ (88–90\%)}$, $Sn\text{ (10–12\%)}$ | Hard, low friction, sonorous, sea-water resistant | Statues, commemorative medals, ship propellers, heavy-duty mechanical bearings, coins | | **German Silver** | $Cu\text{ (50–60\%)}$, $Zn\text{ (20–30\%)}$, $Ni\text{ (10–20\%)}$ | Silvery-white luster, high electrical resistance | Tableware, decorative silverware imitations, electrical rheostats (**Contains 0% Ag!**) | | **Gun Metal** | $Cu\text{ (88\%)}$, $Sn\text{ (10\%)}$, $Zn\text{ (2\%)}$ | Resists high steam pressure and saltwater erosion | Heavy artillery ordnance, high-pressure steam fittings, marine gears | | **Solder (Tanka)** | $Pb\text{ (50\%)}$, $Sn\text{ (50\%)}$ | Low melting point ($\sim 183^\circ\text{C}$), rapid adhesion | Joining electrical circuit components, sealing copper plumbing pipes | | **Stainless Steel** | $Fe\text{ (70–73\%)}$, $Cr\text{ (18\%)}$, $Ni\text{ (8–10\%)}$, $C\text{ (0.1–0.2\%)}$ | Passive

Cr_2O_3 film prevents rust, non-reactive | Surgical instruments, cutlery, kitchen cookware, chemical reaction vessels | | **Nichrome** | Ni\text{ (80\%)}, Cr\text{ (20\%)} | High resistivity, high m.p. (1400^\circ\text{C}), non-oxidizing | Heating elements in electric irons, toasters, water geysers, hair dryers | | **Duralumin** | Al\text{ (95\%)}, Cu\text{ (4\%)}, Mg\text{ (0.5\%)}, Mn\text{ (0.5\%)} | Lightweight, high strength-to-weight ratio | Aircraft fuselage frames, aerospace structural rivets, pressure cookers | | **Alnico** | Fe\text{ (50\%)}, Al\text{ (12\%)}, Ni\text{ (20\%)}, Co\text{ (15–20\%)} | High magnetic coercivity and flux density | Powerful permanent magnets for loudspeakers, electric guitar pickups, MRI scanners | | **Invar** | Fe\text{ (64\%)}, Ni\text{ (36\%)} | Near-zero coefficient of thermal expansion | Surveying measuring tapes, clock pendulums, precision optical instruments | | **Dental Amalgam** | Hg\text{ (50\%)}, Ag\text{ (25–30\%)}, Sn\text{ (10–15\%)}, Cu\text{ (2–5\%)} | Pliable paste that sets into a rock-hard solid | Restorative tooth cavity dental fillings | --- ### Table 3: Types of Chemical Reactions & Everyday Manifestations | Reaction Category | Defining Equation / Archetype | Mechanistic Description | Everyday / Must-Score Example | |---|---|---|---| | **Combination (Synthesis)** | A + B \to AB | Two or more reactants combine to form a single product. Exothermic. | Slaking of quicklime: CaO + H_2O \to Ca(OH)_2 + \text{heat}. Burning of coal: C + O_2 \to CO_2. | | **Decomposition (Thermal)** | AB \xrightarrow{\Delta} A + B | A single compound breaks into multiple simpler substances via heat. | Heating limestone: CaCO_3 \xrightarrow{\Delta} CaO + CO_2 \uparrow. Decomposition of FeSO_4 \cdot 7H_2O. | | **Decomposition (Photolytic)** | 2AB \xrightarrow{h\nu} 2A + B_2 | Chemical breakdown driven by absorption of light photons. | Silver chloride in photography: 2AgCl\text{ (white)} \xrightarrow{h\nu} 2Ag\text{ (grey)} + Cl_2 \uparrow. | | **Single Displacement** | A + BC \to AC + B | A more reactive metal displaces a less reactive metal from its salt solution. | Iron nail in blue vitriol: Fe + CuSO_4\text{ (blue)} \to FeSO_4\text{ (green)} + Cu\text{ (brown deposit)}. | | **Double Displacement (Precipitation)** | AB + CD \to AD\downarrow + CB | Mutual exchange of ions between two aqueous solutions forming a precipitate. | Barium sulphate test: Na_2SO_4\text{(aq)} + BaCl_2\text{(aq)} \to BaSO_4\downarrow\text{ (white ppt)} + 2NaCl\text{(aq)}. | | **Neutralization** | \text{Acid} + \text{Base} \to \text{Salt} + H_2O | H^+ from acid combines with OH^- from base; exothermic (\Delta H = -57.1\text{ kJ/mol}). | Hydrochloric acid with sodium hydroxide: HCl + NaOH \to NaCl + H_2O. | | **Redox Reaction** | \text{Oxidation} + \text{Reduction} | Simultaneous transfer of electrons; one species oxidized, other reduced. | Thermite welding: Fe_2O_3 + 2Al \to Al_2O_3 + 2Fe (Al is oxidized; Fe^{3+} is reduced). |

Detailed Theory Notes

3.1 Physical Properties & Unique Tags of Metals and Non-Metals

1. Mechanical Properties

- **Malleability**: The property that allows a material to be hammered, rolled, or beaten into ultra-thin sheets without fracturing. Gold (Au) is the most malleable element known, followed by Silver (Ag) and Aluminium (Al).
- **Ductility**: The mechanical property of being drawn out into fine longitudinal wires under tensile stress. Gold is the most ductile metal: 1\text{ g} of gold can be drawn into a wire 2\text{ km} long.

- **Hardness vs. Softness**: Most transition metals (Tungsten, Chromium, Iron) exhibit extreme hardness due to overlapping d-orbitals forming strong metallic bonds. In stark contrast, alkali metals (**Sodium, Potassium, Lithium**) have only one valence s-electron and exceptionally large atomic radii, resulting in weak metallic bonding; they can be effortlessly cut with a butter knife.
- **Sonority**: The ability of a metal to produce a deep, ringing acoustic sound when struck, caused by rapid delocalized electron wave dissipation. Used in bells and musical wires (Bronze, Bell Metal).

2. Thermal and Electrical Conductivity

- **Conduction Mechanism**: Metals contain a "sea" of free, delocalized conduction electrons. When an electric potential or temperature gradient is applied, these electrons undergo directional drift, carrying thermal and electrical energy at high velocities.
- **Conductivity Sequence**: \text{Electrical Conductivity: } Ag > Cu > Au > Al > W > Fe Although Silver is the superior conductor, **Copper and Aluminium** are preferred for domestic and industrial wiring due to cost efficiency and lower density.
- **Thermal Anomalies**: Lead (Pb) and Mercury (Hg) have the lowest thermal conductivity among common metals, while Diamond (a non-metal) has a thermal conductivity 4–5 times higher than Copper due to phonon lattice vibrations.

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Densest Element:   Osmium (Os, 22.59 g/cm³)
Lightest Metal:    Lithium (Li, 0.534 g/cm³)
Liquid Metal:      Mercury (Hg, m.p. -38.8°C)
Liquid Non-Metal:  Bromine (Br₂, reddish-brown)
Highest m.p. Metal: Tungsten (W, 3422°C)
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3.2 Carbon Allotropes & Thermodynamic Stability

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                                  ┌── Diamond: sp³, tetrahedral 3D network; hardest; insulator;
                                  ├── Graphite: sp², planar hexagonal layers; electrical conduc
Carbon Allotropes (Pure Carbon) ──┤
                                  ├── Fullerenes: C₆₀ buckyball; 20 hexagons + 12 pentagons; ca
                                  └── Graphene: Single 2D atomic sheet; ballistic conductivity;
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1. Diamond

- **Structure**: Each carbon atom is sp^3 hybridized and covalently bonded to 4 neighboring carbon atoms in a rigid, three-dimensional tetrahedral lattice with a uniform C-C bond length of 1.54\text{ \AA}.
- **Properties**: Hardest known natural substance (10 on Mohs hardness scale); density 3.51\text{ g/cm}^3; high refractive index (2.42) and small critical angle (24.4^\circ), causing brilliant total internal reflection.
- **Electrical & Thermal Behavior**: Because all 4 valence electrons are tightly locked in localized covalent \sigma-bonds, diamond has no free electrons and is a **complete electrical insulator**. However, its rigid lattice allows high-frequency phonon vibrations, making it an **exceptional conductor of heat**.

2. Graphite

- **Structure**: Each carbon atom is sp^2 hybridized and covalently bonded to 3 neighboring carbon atoms in a flat hexagonal planar ring (C-C bond length 1.42\text{ \AA}). These planar layers are stacked parallel to one another at a distance of 3.35\text{ \AA} and held together only by weak van der Waals dispersion forces.
- **Properties**: Soft, greasy, and slippery; layers slide over one another under shear stress, making graphite an ideal **solid dry lubricant** for high-temperature machinery where liquid oil would decompose.
- **Electrical Conduction**: The fourth valence electron of each carbon atom occupies an unhybridized p-orbital that overlaps sideways to form delocalized \pi-bonds across the hexagonal sheets. These free electrons move along the sheets, making graphite a **good conductor of electricity** (used as electrodes in dry cells and electric arc furnaces).
- **Thermodynamic Ground State**: In chemical thermodynamics, **Graphite is the most stable allotrope of carbon** under standard ambient conditions (298.15\text{ K}, 1\text{ bar}). By international IUPAC convention, the standard molar enthalpy of formation of graphite is defined as zero (\Delta_f H^\circ\text{ [graphite]} = 0\text{ kJ/mol}), whereas for diamond it is +1.9\text{ kJ/mol}. Diamond is technically thermodynamically metastable and converts spontaneously—though infinitely slowly—into graphite.

3. Fullerenes & Graphene

- **Buckminsterfullerene (C_{60})**: Discovered in 1985 by Harold Kroto, Robert Curl, and Richard Smalley (Nobel Prize in Chemistry, 1996). Contains 60 carbon atoms organized in a spherical geodesic cage consisting of **20 hexagonal rings and 12 pentagonal rings**. No two pentagons share an edge (Isolated Pentagon Rule).
- **Graphene**: Isolated in 2004 by Andre Geim and Konstantin Novoselov at the University of Manchester using mechanical exfoliation ("Scotch tape method", Nobel Prize in Physics, 2010). Graphene is a single atom-thick two-dimensional sheet of sp^2 carbon atoms arranged in a honeycomb lattice with ballistic electronic mobility, extreme thermal conductivity, and tensile strength over 130\text{ GPa}.

3.3 The Reactivity Series & Chemical Behavior of Metals

The **Electrochemical / Activity Series** ranks metals in descending order of their standard oxidation potentials (tendency to lose electrons in aqueous solution):

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K  >  Na  >  Ca  >  Mg  >  Al  >  Zn  >  Fe  >  Pb  >  [H]  >  Cu  >  Hg  >  Ag  >  Pt  >  Au
<---------------- High Reactivity ---------------->   <-------------- Noble Metals -----------
Displaces H₂ from cold water                         Displaces H₂ from steam   Cannot displace
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1. Reaction with Water

- **Cold Water**: Only the most electropositive alkali and alkaline earth metals (K, Na, Ca) react with cold water: 2Na + 2H_2O \to 2NaOH + H_2 \uparrow + \text{heat} Sodium and Potassium float, melt into silvery spheres, and skitter across the surface; the liberated H_2 gas catches fire with explosive popping. Calcium reacts less violently, and the clinging H_2 bubbles cause the metal to float.

- **Hot Water**: Magnesium reacts with hot water to form $Mg(OH)_2$ and H_2.
- **Steam**: Aluminium, Zinc, and Iron do not react with cold or hot liquid water, but react with steam at red heat: $2Al + 3H_2O\text{ (steam)} \to Al_2O_3 + 3H_2 \uparrow$ $3Fe + 4H_2O\text{ (steam)} \rightleftharpoons Fe_3O_4 + 4H_2 \uparrow$
- **Unreactive**: Pb, Cu, Hg, Ag, Au do not react with water or steam at any temperature.

2. Reaction with Acids

- Metals situated above Hydrogen in the reactivity series displace hydrogen gas from dilute hydrochloric and sulphuric acids: $Zn + 2HCl \to ZnCl_2 + H_2 \uparrow$ $Fe + H_2SO_4 \to FeSO_4 + H_2 \uparrow$
- **Exception for Nitric Acid (HNO_3)**: Concentrated and moderately dilute HNO_3 is an aggressive oxidizing agent. It oxidizes evolved hydrogen immediately into water, reducing its own nitrogen species: $3Cu + 8HNO_3\text{ (dilute)} \to 3Cu(NO_3)_2 + 2NO\text{ (Nitric oxide)} + 4H_2O$ $Cu + 4HNO_3\text{ (conc.)} \to Cu(NO_3)_2 + 2NO_2\text{ (Nitrogen dioxide)} + 2H_2O$ Only **Magnesium (Mg)** and **Manganese (Mn)** react with very dilute (1%) cold HNO_3 to evolve pure H_2 gas: $Mg + 2HNO_3\text{ (1\% dilute)} \to Mg(NO_3)_2 + H_2 \uparrow$

3. Aqua Regia Chemistry

- Aqua Regia is a mixture of **$3\text{ parts concentrated } HCl$ and $1\text{ part concentrated } HNO_3$** by volume (3:1 ratio).
- **Mechanism of Dissolution**: $HNO_3 + 3HCl \to NOCl\text{ (Nitrosyl chloride)} + Cl_2\text{ (free nascent chlorine)} + 2H_2O$ The highly reactive nascent chlorine attacks noble metals that resist all individual acids, forming soluble coordinate complexes: $Au + 3[Cl] + HCl \to H[AuCl_4]\text{ (Chloroauric acid)}$ $Pt + 4[Cl] + 2HCl \to H_2[PtCl_6]\text{ (Chloroplatinic acid)}$

3.4 Important Ores, Extraction Processes & Metallurgy

1. Extractive Steps in Metallurgy

1. **Crushing and Pulverization**: Mechanical reduction of large rocky boulders into fine powder using jaw crushers and ball mills.
2. **Concentration / Beneficiation**: Removal of unwanted rocky, earthy gangue impurities: - *Gravity Separation (Hydraulic Washing)*: Relies on density differences (used for heavy oxide ores like haematite Fe_2O_3). - *Magnetic Separation*: Uses electromagnetic rollers (separates magnetic wolframite from non-magnetic tin stone SnO_2). - *Froth Flotation*: Used selectively for **sulphide ores** (ZnS, PbS, $CuFeS_2$). Pulverized ore is mixed with water, pine oil (collector), and sodium ethyl xanthate. Air is vigorously bubbled through; oil-coated sulphide particles adhere to the rising froth, while hydrophilic gangue sinks to the tank bottom. - *Chemical Leaching*: Selective dissolution of metal in a chemical reagent (e.g., Bayer's process for bauxite using hot concentrated NaOH).
3. **Conversion to Oxide (Calcination vs. Roasting)**: - *Calcination*: Heating carbonate or hydrated ores below melting point in the **absence/deficiency of air** to expel moisture and CO_2 ($CaCO_3 \to CaO + CO_2$). - *Roasting*: Heating sulphide ores below melting point in **excess air** to oxidize sulphides into oxides and volatile SO_2 gas ($2PbS + 3O_2 \to 2PbO + 2SO_2$).
4. **Reduction of Metal Oxide to Free Metal**: - *Smelting with Carbon (Coke)*: Used for moderate-reactivity metals (Fe, Zn, Pb, Sn). - *Thermite Reduction (Aluminothermy)*: Reduction of refractory

oxides (Cr_2O_3, Mn_3O_4, Fe_2O_3) using aluminium powder. Highly exothermic ($>2500^\circ\text{C}$); used for in-situ railway track welding. - *Electrolytic Reduction*: Used for highly electropositive metals (Na, K, Ca, Mg, Al) whose oxides cannot be reduced by carbon.

5. **Refining / Purification**: - *Liquation*: For low-melting metals like Tin (Sn) and Bismuth (Bi). - *Distillation*: For volatile metals with low boiling points like Zinc (Zn) and Mercury (Hg). - *Electrolytic Refining*: Impure metal forms anode, pure metal strip forms cathode, metal salt is electrolyte (used for Cu, Ag, Au, Al). - *Zone Refining*: Based on the principle that impurities are more soluble in the molten melt than in the solid metal; yields ultra-pure semiconductors (Silicon Si, Germanium Ge). - *Vapour Phase Refining*: Mond process for Nickel ($Ni + 4CO \xrightarrow{50^\circ\text{C}} Ni(CO)_4 \xrightarrow{200^\circ\text{C}} Ni + 4CO$); Van Arkel-de Boer process for Titanium (Ti) and Zirconium (Zr) using iodine.

3.5 Corrosion Mechanisms & Protective Technologies

1. Electrochemical Theory of Rusting

Rusting of iron is not a simple chemical combination, but an **electrochemical cell (galvanic corrosion)** set up on the microscopic metal surface: - **Anodic Region**: Iron metal oxidizes, releasing electrons: $\text{Anode: } Fe\text{ (s)} \to Fe^{2+}\text{ (aq)} + 2e^- \quad (E^\circ = -0.44\text{ V})$ - **Cathodic Region**: Electrons migrate through the metal to a separate region with dissolved oxygen and moisture: $\text{Cathode: } O_2\text{ (g)} + 4H^+\text{ (aq)} + 4e^- \to 2H_2O\text{ (l)} \quad (E^\circ = +1.23\text{ V})$ - **Overall Cell Reaction**: $2Fe + O_2 + 4H^+ \to 2Fe^{2+} + 2H_2O$ Ferrous ions are further oxidized by atmospheric oxygen to ferric state and precipitate as hydrated ferric oxide: $4Fe^{2+} + O_2 + 4H_2O + 2xH_2O \to 2Fe_2O_3 \cdot xH_2O\text{ (Rust)} + 8H^+$ - **Weight Consequence**: Because atmospheric oxygen and water molecules are chemically bonded into the crystal structure of the iron object, the **total weight of the iron article increases upon rusting**.

2. Corrosion in Other Metals

- **Copper Patina**: When exposed to moist air containing CO_2 and SO_2 over long periods, copper develops a protective green surface layer consisting of **basic copper carbonate**: $2Cu + O_2 + CO_2 + H_2O \to CuCO_3 \cdot Cu(OH)_2\text{ (Green Patina)}$ (Seen on the Statue of Liberty and historic temple domes).
- **Silver Tarnishing**: Silver is inert to oxygen, but reacts with trace hydrogen sulphide (H_2S) in polluted air to form a black surface tarnish of **silver sulphide**: $2Ag + H_2S \to Ag_2S\text{ (Black tarnish)} + H_2$

3. Prevention Strategies

- **Galvanization**: Dipping iron articles into a bath of molten Zinc ($\approx 450^\circ\text{C}$). Zinc oxidizes preferentially to form a tough film of zinc hydroxycarbonate. Crucially, because Zinc is more electropositive than Iron ($E^\circ_{Zn^{2+}/Zn} = -0.76\text{ V}$ vs $E^\circ_{Fe^{2+}/Fe} = -0.44\text{ V}$), zinc acts as a **sacrificial anode**. Even if the zinc coating is scratched, zinc continues to oxidize and supply electrons to the iron, preventing the iron from rusting.
- **Cathodic Protection (Sacrificial Anode)**: Underground steel pipelines and ship hulls are connected via electrical cables to blocks of more reactive sacrificial metals like **Magnesium (Mg) or Zinc (Zn)**.

The sacrificial metal corrodes over time and is periodically replaced, preserving the steel structure indefinitely.

- **Anodizing**: An electrolytic oxidation process applied primarily to aluminium articles. The aluminium part forms the anode in an acid bath (H_2SO_4); oxygen liberated at the anode reacts with aluminium to produce a thick, durable, corrosion-resistant surface layer of **aluminium oxide (Al_2O_3)**, which can be dyed in bright decorative colors.

Bilingual Terminology -- Metals and Chemical Reactions

English Term	Hindi Term	Key Anchor
Metal (धातु)	धातु	Lustrous, malleable, ductile, good conductor; loses electrons
Non-metal (अधातु)	अधातु	Dull, brittle, poor conductor (except graphite); gains electrons
Alloy (मिश्र धातु)	मिश्र धातु	Mixture of two or more metals (or metal + non-metal)
Corrosion (संक्षारण)	संक्षारण	Deterioration of metals by reaction with environment; e.g., rusting
Rusting (जंग लगना)	जंग लगना	Corrosion of iron; Fe2O3.xH2O (hydrated iron oxide); needs O2 + H2O
Oxidation (ऑक्सीकरण)	ऑक्सीकरण	Loss of electrons; increase in oxidation state
Reduction (अपचयन)	अपचयन	Gain of electrons; decrease in oxidation state
Redox Reaction (ऑक्सीकरण-अपचयन)	रेडॉक्स अभिक्रिया	Simultaneous oxidation and reduction in same reaction
Calcination (निस्तापन)	निस्तापन	Heating ore in limited air; e.g., CaCO3 -> CaO + CO2
Roasting (भर्जन)	भर्जन	Heating sulphide ore in excess air; converts to oxide; e.g., ZnS + O2 -> ZnO + SO2
Smelting (प्रगलन)	प्रगलन	Melting ore with reducing agent (coke) to extract metal
Electrolytic Refining (विद्युत-अपघटनी परिष्करण)	विद्युतअपघटनी शोधन	Purification of metals using electrolysis; crude as anode, pure as cathode
Thermite Reaction	थर्माइट अभिक्रिया	Al + Fe2O3 -> Al2O3 + Fe + huge heat; welding railway tracks
Galvanisation (जस्तीकरण)	गैल्वनीकरण	Coating iron with zinc to prevent rusting
Anodising (एनोडाइजिंग)	एनोडाइजिंग	Electrochemical coating of aluminium with Al2O3 (protection layer)
Reactivity Series (क्रियाशीलता श्रृंखला)	अभिक्रियाशीलता क्रम	Metals arranged by reactivity: K>Na>Ca>Mg>Al>Zn>Fe>Ni>Sn>Pb>H>Cu>Hg>Ag>Au>Pt

One-Liner Revision -- Metals and Chemical Reactions

#	Fast Fact
1	Most reactive metal = **Potassium (K)**; reacts violently with water
2	Least reactive metal = **Platinum (Pt)** / Gold (Au) = noble metals
3	Most malleable metal = **Gold (Au)** (1g can be beaten into 1m2 sheet)
4	Best conductor of electricity = **Silver (Ag)**; then Cu, then Au, then Al
5	Hardest metal = **Chromium (Cr)**; hardest element = Carbon (diamond)
6	Lightest metal = **Lithium (Li)**; floats on water and oil
7	Mercury (Hg) = only metal that is liquid at room temperature
8	Gallium (Ga) = melts in palm (melting point 29.76 degC)
9	Stainless steel = Fe + Cr (10.5%+) + Ni; rust-resistant
10	Brass = **Cu + Zn**; Bronze = Cu + Sn; Gunmetal = Cu + Sn + Zn
11	German silver / Nickel silver = Cu + Zn + Ni (no silver!)
12	Solder = **Pb + Sn** (used in electronics for joining); being replaced by Sn-Ag-Cu
13	Bell metal = **Cu + Sn** (high tin content); used for bells and cymbals
14	Duralumin = **Al + Cu + Mg + Mn**; lightweight strong alloy; aircraft construction
15	Rusting requires both **water** and **oxygen**; prevented by galvanisation, painting, oiling
16	Galvanisation = coating iron with **zinc (Zn)**; zinc acts as sacrificial anode
17	Thermite reaction: 2Al + Fe2O3 -> 2Fe + Al2O3 + heat (~2500 degC); railway welding
18	Reactivity series: metals above H displace H from acids; K, Na, Ca react with cold water
19	Gold does not corrode; found native (in free state) in nature
20	Bauxite (Al2O3.2H2O) = ore of aluminium; Haematite (Fe2O3) = ore of iron
21	Bayer Process = extraction of alumina from bauxite; then Hall-Heroult for Al electrolysis
22	Blast furnace = extraction of iron from haematite using coke + limestone
23	Pig iron = crude iron from blast furnace; high carbon (4-5%); brittle
24	Wrought iron = nearly pure iron; very low carbon (<0.02%); most malleable
25	Steel = iron + carbon (0.2-2%); iron + Cr = stainless; iron + Mn = manganese steel
26	Amalgamation = extraction of Ag/Au using Hg (forms amalgam)
27	Electrolytic refining: crude metal at anode (-) dissolves; pure at cathode (-) deposits
28	Patina = green coating on copper = basic copper carbonate [Cu(OH)2.CuCO3]

#	Fast Fact
29	Verdigris (patina on Statue of Liberty) = green due to copper oxidation
30	Chemical change = new substance formed (rust); Physical change = no new substance (melting)